

10-Month Master Roadmap: AI/ML, DSA, & System Design for MNCs

Goal: Transition to a high-paying MNC role (Java Developer/ML Engineer) by March 2027.
Current Status: Starting from scratch with 10 months of dedicated preparation.

I. The Unified Tech Stack

- **Core Engineering (Java):** Java 21+, Spring Boot, Hibernate, JUnit (for Testing/Quality).
- **Data Structures & Algorithms:** Java Collections Framework, LeetCode (Focus: Medium/Hard).
- **AI & Machine Learning (Python):** Python 3.12+, NumPy, Pandas, Scikit-Learn, PyTorch, Hugging Face (GenAI).
- **System Design & Databases:** PostgreSQL (SQL), MongoDB (NoSQL), Redis (Caching), Kafka (Messaging).
- **Cloud & DevOps:** AWS (EC2, S3, Lambda, SageMaker), Docker, Kubernetes.

II. 10-Month Execution Timeline

Phase	Timeline	Primary Focus	Key Milestone
Phase 1: Foundation	Month 1 - 2	Java Deep Dive + Basic DSA (Arrays, Strings, Linked Lists).	Solve 50+ LeetCode Easy/Medium problems in Java.
Phase 2: System Architecture	Month 3 - 4	Spring Boot Backend + Low Level Design (LLD) + SQL.	Build a REST API with Java and PostgreSQL.
Phase 3: AI/ML Core	Month 5 - 7	Python for Data Science + ML Math + Scikit-Learn.	Kaggle Competition entry & 3 ML Model Projects.

Phase	Timeline	Primary Focus	Key Milestone
Phase 4: Cloud & Scalability	Month 8 - 9	High Level Design (HLD) + AWS + Docker + Kubernetes.	AWS Cloud Practitioner knowledge + Dockerized Apps.
Phase 5: Integration & Jobs	Month 10	Full-stack AI Project + Resume + Mock Interviews.	Deployed End-to-End AI System on AWS.

III. Learning Priority (The "What to Learn First" Rule)

1. **DSA (Always concurrent):** Start this on Day 1 and do not stop until you get the job.
2. **Java/Backend:** Learn this first because it leverages your current background and provides immediate "Job Readiness" for standard Developer roles.
3. **AI/ML:** Learn this after you understand how to build a basic system; it is easier to "plug in" AI to a working system than to build an AI model without knowing how to deploy it.
4. **System Design & Cloud:** These are the "Senior" skills that justify a high salary. Learn these once you are comfortable with coding and models.

IV. Weekly Commitment Strategy

- **Weekday (1 Hours):** practice coding and maintain github streak.
- **Weekend (5-6 Hours):** Project Building, Deep-dive into Cloud/Architecture, and Weekly Review.